

RECURSIVE

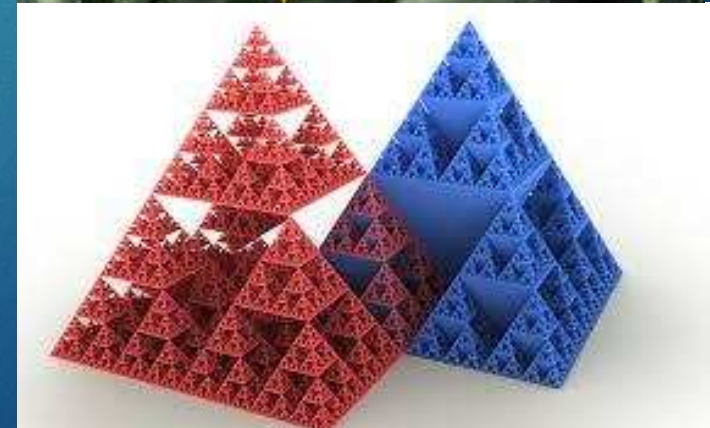
DATA STRUCTURE

DEFINITION

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- The solution of a big problem depends on solutions to smaller instances of the same problem.



HOW TO SOLVE PROBLEM BY RECURSIVE

- ▶ if problem is “small enough”
- ▶ solve it directly
- ▶ else
- ▶ break into one or more smaller sub-problems
- ▶ solve each sub-problem recursively
- ▶ combine results into solution to whole problem

BASIC EXERCISES

RECURSIVE

FACTORIAL

BASIC RECURSIVE EXERCISES

FACTORIAL DEFINITION

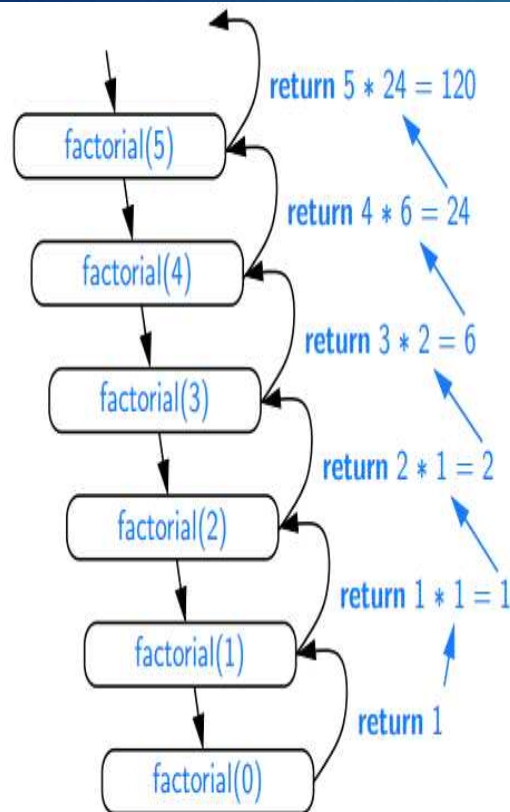
- $0! = 1$
- $1! = 1 * 1$
- $2! = 1 * 2$
- $3! = 1 * 2 * 3$
- ...
- $N! = 1 * 2 * 3... * N$

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1 & \text{if } n \geq 1. \end{cases}$$

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot (n-1)! & \text{if } n \geq 1. \end{cases}$$

FACTORIAL IMPLEMENT

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```
public static int factorial(int n) throws IllegalArgumentException {  
    if (n < 0)  
        throw new IllegalArgumentException(); // argument must be nonnegative  
    else if (n == 0)  
        return 1; // base case  
    else  
        return n * factorial(n-1); // recursive case  
}
```

Figure 5.1: A recursion trace for the call factorial(5).

DECIMAL TO BINARY

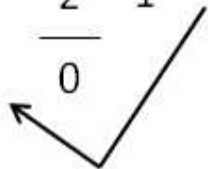
BASIC RECURSIVE EXERCISES

DECIMAL TO BINARY EXAMPLE

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Decimal

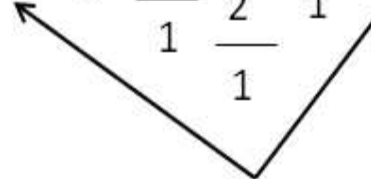
$$\begin{array}{r|l} 2 & 2 \\ \hline 2 & 1 \\ \hline 0 & \end{array}$$


→

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Binary

Decimal

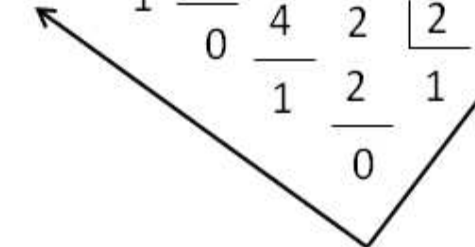
$$\begin{array}{r|l} 15 & 2 \\ \hline 14 & 7 & 2 \\ \hline 1 & 6 & 3 & 2 \\ \hline & 1 & 2 & 1 \\ & & 1 & \end{array}$$


→

1111

Binary

Decimal

$$\begin{array}{r|l} 43 & 2 \\ \hline 42 & 21 & 2 \\ \hline 1 & 20 & 10 & 2 \\ \hline & 1 & 10 & 5 & 2 \\ & & 0 & 4 & 2 & 2 \\ & & & 1 & 2 & 1 \\ & & & & 0 & \end{array}$$


→

101011

Binary

ENGLISH RULER

BASIC RECURSIVE EXERCISES

DRAWING AN ENGLISH RULER

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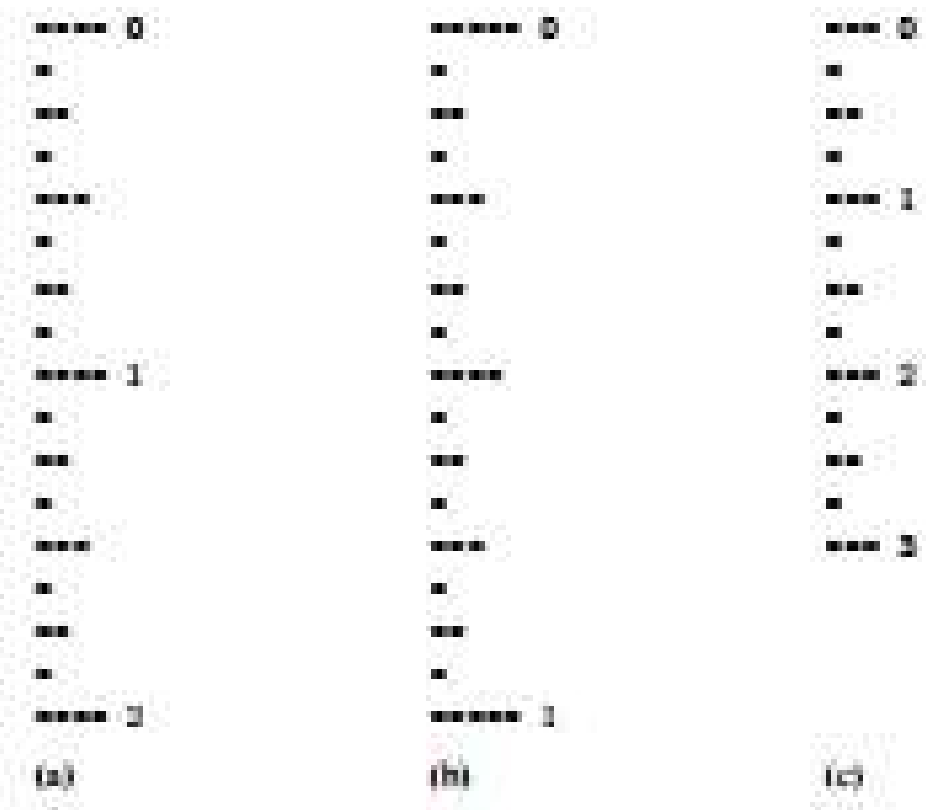


Figure 5.2: Three sample outputs of an English ruler drawing: (a) a 2-inch ruler with major tick length 4; (b) a 1-inch ruler with major tick length 5; (c) a 3-inch ruler with major tick length 3.

DRAWING AN ENGLISH RULER

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```
/* Draws an English ruler for the given number of inches and major tick length. */
public static void drawRuler(int inches, int majorLength) {
    drawLine(majorLength, 0);           // draw inch 0 line and label
    for (int j = 1; j <= inches; j++) {
        drawInterval(majorLength - 1); // draw interior ticks for inch
        drawLine(majorLength, j);      // draw inch j line and label
    }
}

private static void drawInterval(int centrallength) {
    if (centrallength >= 1) {           // otherwise, do nothing
        drawInterval(centrallength - 1); // recursively draw top interval
        drawLine(centrallength);         // draw center tick line (without label)
        drawInterval(centrallength - 1); // recursively draw bottom interval
    }
}

private static void drawLine(int tickLength, int tickLabel) {
    for (int j = 0; j < tickLength; j++)
        System.out.print("-");
    if (tickLabel >= 0)
        System.out.print(" " + tickLabel);
    System.out.print("\n");
}

/* Draws a line with the given tick length (but no label). */
private static void drawLine(int tickLength) {
    drawLine(tickLength, -1);
}
```

HA NOI TOWER

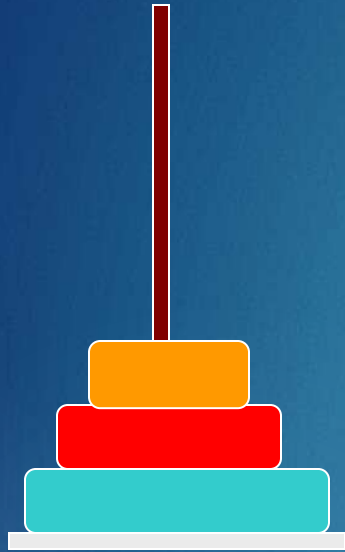
RULE OF HANOI TOWER GAME

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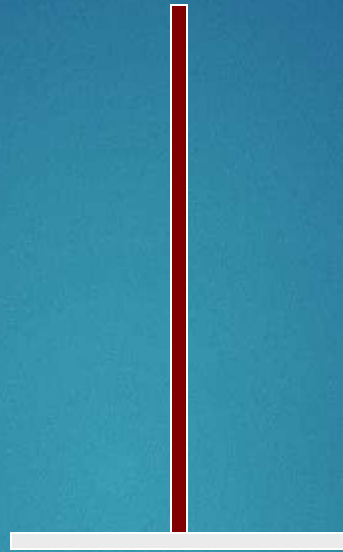
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1. Only one disk can be moved at a time.
2. Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack.
3. No disk may be placed on top of a smaller disk.

HANOI TOWER GAME



A



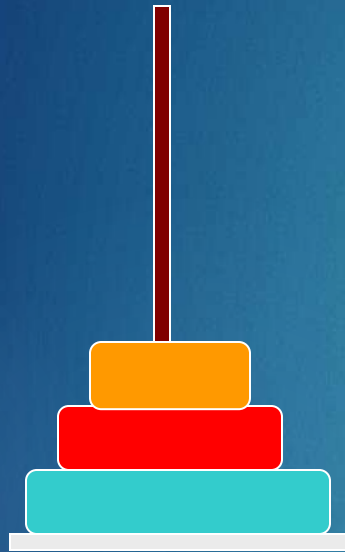
B



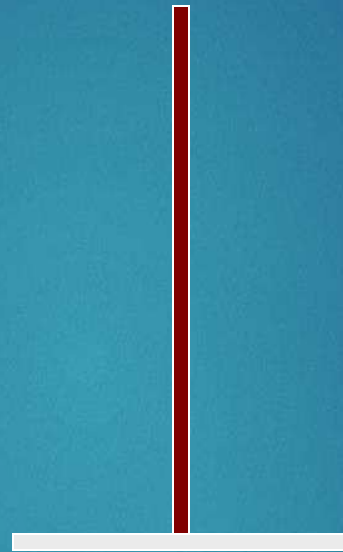
C

How to move 3 disks from A column to C column with minimum of steps?

SOLUTION



A



B



C

- The minimum number of moves required to solve a Tower of Hanoi puzzle is $2^n - 1$, where n is the number of disks.

Recursive solution

- RULE:
 - label 3 columns **begin(A), middle(B), end(C)**
 - let n be the total number of discs
 - number the discs from 1 (smallest, topmost) to n (largest, bottommost)
 - To move n discs from **begin** column to **end** column
- RECURSIVE SOLVE:
 - Recursive move n-1 discs from begin to end. With **begin** not change, **end, middle (swap end and middle column)**
 - move disc n from **begin** to **end**
 - Recursive move n-1 discs from begin to end. With **middle, begin, end** not change (**swap begin and middle column**)

Tower of Hanoi Recursive Algorithm:

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- ▶ **Function TowersofHanoi(n ,start, mid, end){**
//begin start A, mid B, end C
- ▶ N = number of disks

If N == 1
Move Single disk from A to C
- ▶ If N > 1
Move n-1 disks from start A to B
TowersofHanoi(n-1,start, end , mid)
- ▶ Move last Disk from A to C
- ▶ Move n-1 disks from B to C.
TowersofHanoi(n-1,start, mid, end)

FIBONACCI

FIBONACCI IN NATURE

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FIBONACCI IN NATURE

21



FIBONACCI

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1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

$$F(n) := \begin{cases} 1, & \text{khi } n = 1; \\ 1, & \text{khi } n = 2; \\ F(n-1) + F(n-2) & \text{khi } n > 2. \end{cases}$$

PASCAL'S TRIANGLE

PASCAL'S TRIANGLE

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1										
1		1								
1			2							
1				3						
1					4					
1						5				
1							6			
1								7		
1									8	
1										1

$$(a+b)^0 = 1$$

$$(a+b)^1 = a+b$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

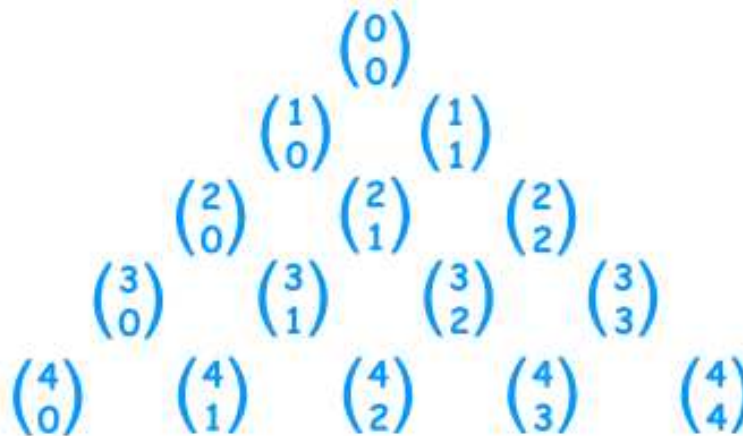
$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

PASCAL'S TRIANGLE

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It is commonly called "n choose k" and written like this: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

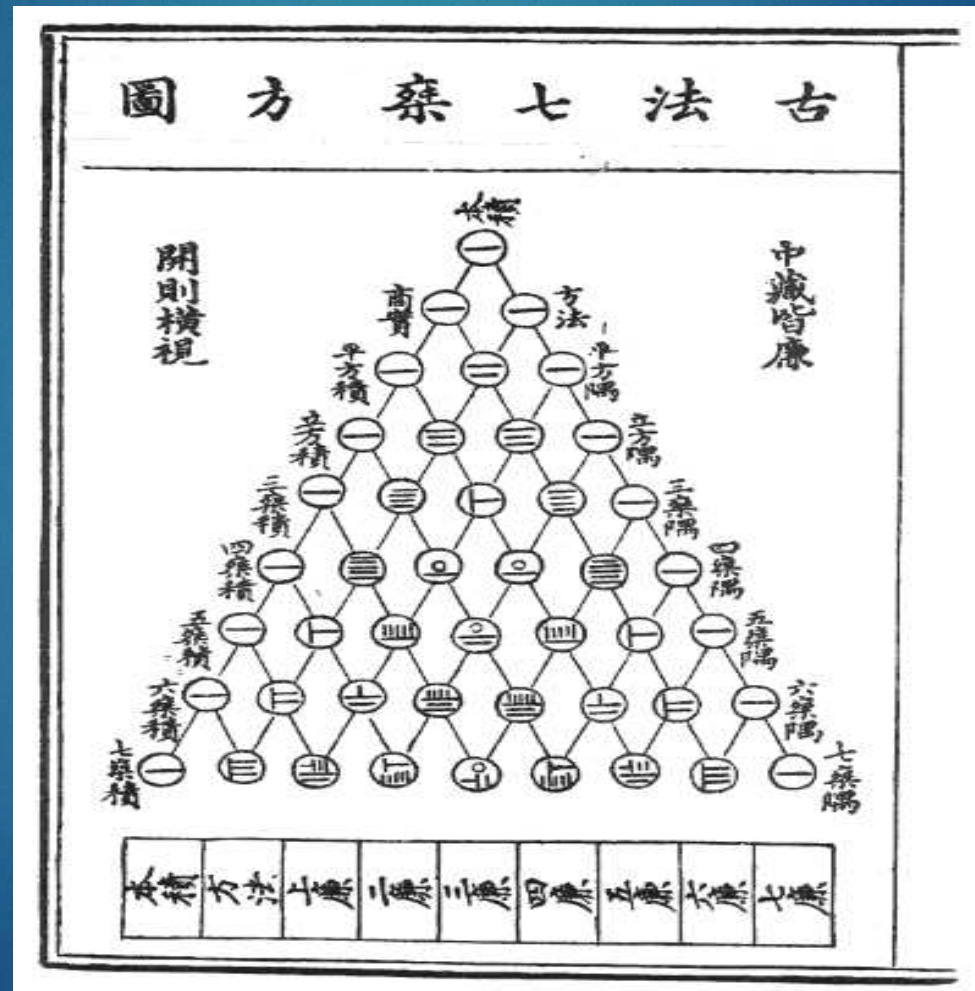
Notation: "n choose k" can also be written $C(n,k)$, nC_k or even ${}_nC_k$.



Chu Shi-Chieh's book "Ssu Yuan Yü Chien" - 1303

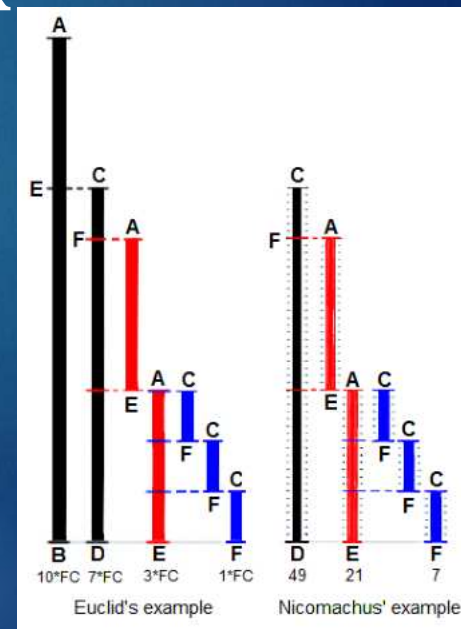
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GREATEST COMMON DIVISOR (ước chung lớn nhất)

EUCLIDEAN ALGORITHM



Example

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► $\text{GCD}(42, 56) = 14$

$$\frac{42}{56} = \frac{3 \cdot 14}{4 \cdot 14} = \frac{3}{4}.$$

Euclidean algorithm

- ▶ With $a, b \in \mathbb{N}$
- ▶ Condition $a \geq b, a, b \neq 0$
- ▶ Has $a = b \cdot q + r$ (with $r = a \bmod b$)

$$\text{UCLN}(a, b) = \begin{cases} b & \text{nếu } r = 0 \\ \text{UCLN}(b, r) & \text{nếu } r \neq 0 \end{cases}$$

TRY TO SOLVE SOME ALGEBRA PROBLEMS

ALGEBRA PROBLEMS

1. $S(n) = 1 - 2 + 3 - 4 + \dots + ((-1)^{(n+1)}) \cdot n, n > 0$
2. $S(n) = 1 + 1 \cdot 2 + 1 \cdot 2 \cdot 3 + \dots + 1 \cdot 2 \cdot 3 \dots n, n > 0$
3. $S(n) = 1^2 + 2^2 + 3^2 + \dots + n^2, n > 0$
4. $S(n) = 1 + 1/2 + 1/(2 \cdot 4) + 1/(2 \cdot 4 \cdot 6) + \dots$
 $+ 1/(2 \cdot 4 \cdot 6 \cdot 2n), n > 0$